**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS) Regular EXAMINATION October 2025**
**THIRD SEMESTER**
**MG3VACTXLA04 – TEXTILE PROCESSING FOR SUSTAINABILITY**
**ANSWER KEY**
**Part – A**
**MCQ**
**Answer all questions**
**Each question carries 1 mark**

1. Recognize the method that reduces transportation-related emissions.
A. Local production
2. Which biodegradable textile is recognized as a fast-growing renewable resource?
A. Bamboo
3. Which of the following is NOT a benefit of recycling and upcycling in textiles?
A. Increased landfill use
4. The method of designing clothes so that fabric waste is minimized is called:
A. Zero waste pattern cutting
5. Recognize the sustainability dimension that increases brand reputation and loyalty.
A. Social responsibility
6. Printed textiles used for branding and marketing come under.
A. Advertising and promotion
7. Which dyeing method is ideal for large- scale continuous production?
A. Continuous Dyeing
8. Which of the following in not a benefit of natural dyeing?
A. Low colour fastness
9. Dyeing plays a major role in textile because:
A. All of the above
10. Natural dyes are considered sustainable because they are:
A. Biodegradable and renewable

**Part B**
**Fill in the blanks**
**Answer all questions**
**Each question carries 1 mark**

11. Printing detailed, complex designs with high precision is a key advantage of _____ printing.
A. Digital printing
12. Sustainability in textiles aims to minimize _____social, and economic impacts.
A. Environmental
13. Promoting _____, human rights, and community well-being is the focus of social sustainability.
A. Fair labour practices
14. Sustainable textile production can create new job opportunities in areas like _____ and upcycling.
A. Recycling
15. Waste reduction in textile production helps lower _____pollution.
A. Environmental
16. _____cotton is grown without toxic pesticides or synthetic fertilizers.
A. Organic
17. The natural origin of dyes makes them _____and _____
A. Biodegradable and ecofriendly
18. Dyeing fibers before they are spun in to yarn is called_____ dyeing.
A. Stock dyeing
19. In block printing _____ blocks are curved with designs.

A. Wooden

20. Shade _____is a common problem in natural dyeing.

A. Variation

**Part C**

**One Word Questions**

**Answer all questions**

**Each question carries 1 mark**

21. What property of nano technology textiles protects skin from sunlight?

A. UV protection

22. Which type of cotton is most commonly used in sustainable fashion?

A. Organic cotton

23. What do we call making bags or accessories from old clothes?

A. Upcycling

24. What certification evaluates recyclability and non-toxicity of textile products?

A. GRS (Global Recycled Standard)

25. What practice involves transforming waste materials into better-quality products?

A. Upcycling

26. Identify the practice that prevents child forced labour.

A. Fair trade

27. What are the characteristics of natural dyes?

A. Biodegradability

28. Which factor controls how deeply the dye enters the fabric?

A. Dye penetration

29. What aspect of eco-friendly fashion focuses on honesty in supply chains?

A. Transparency

30. State the printing method uses hand-carved blocks.

A. Block printing

**Part D**

**Short Answer Questions**

**Answer any 5 questions**

**Each question carries 2 marks**

31. Recall how dyeing in textiles leads to water pollution.

A. Dyeing in textiles leads to water pollution through the release of untreated dye effluent, containing chemicals, heavy metals, and synthetic dyes, which contaminate water sources and harm aquatic life.

32. What are the major challenges in adopting waterless dyeing technologies?

A. Major challenges include high initial investment costs, limited scalability, and potential performance trade-offs, such as colourfastness and fabric quality, hindering widespread adoption of waterless dyeing technologies.

33. Define sustainable textile finishing.

A. Sustainable textile finishing refers to processes that minimize environmental impact, using eco-friendly chemicals, reducing water and energy consumption, and promoting biodegradability while maintaining fabric performance and quality.

34. What happens during the dyeing stage?

A. During the dyeing stage, fibres or fabrics are immersed in a dye bath, where the dye molecules penetrate and bond with the material, imparting colour. Chemicals and fixatives may be added to enhance colourfastness and uniformity.

35. How does dyeing contribute to the functionality of textiles?

A. Dyeing enhances textiles' aesthetic appeal and can also impart functional properties like UV protection, antimicrobial effects, and thermal regulation, depending on the type of dye and additives used.

36. What is the role of responsible Sourcing in eco-friendly fashion?

A. Responsible sourcing in eco-friendly fashion ensures that materials are obtained in a way that minimizes environmental harm and promotes fair labour practices. It involves selecting sustainable materials, reducing waste, and partnering with suppliers who adhere to strict environmental and social standards, ultimately reducing the fashion industry's ecological footprint.

**Part E**

**Long Answer Questions**

**Answer any 2 questions**

**Each question carries 5 marks**

37. Discuss the various purposes of textile printing and explain how it serves both decorative and functional roles.

A. Textile printing serves multiple purposes, blending decorative and functional roles. Decoratively, it adds visual appeal, creates patterns, and enhances aesthetics. Functionally, printing can impart properties like water resistance, antimicrobial effects, or thermal insulation. Techniques like discharge printing, pigment printing, and digital printing allow for diverse applications, from fashion statements to performance textiles, making textile printing a versatile tool for both beauty and functionality.

38. Explain how each of the 4 R's Reduce, Reuse, Recycle, and Recover contribute to sustainability.

The 4 Rs - Reduce, Reuse, Recycle, and Recover - are key strategies for sustainability.

- Reduce: Minimizes waste generation and resource consumption, decreasing the environmental footprint.
- Reuse: Extends product lifespan, reducing the need for new raw materials and waste generation.
- Recycle: Transforms waste into new products, conserving resources and reducing landfill waste.
- Recover: Involves recovering materials or energy from waste, reducing landfill disposal and extracting value from discarded materials.

Together, the 4 Rs help conserve resources, reduce pollution, and promote a circular economy, significantly contributing to sustainability.

39. How do nanotech and enzyme-based processing aid sustainable textiles?

Nanotechnology and plasma-based processing contribute to sustainable textile production by:

- Reducing chemical usage and water consumption
- Enhancing fabric properties like durability, stain resistance, and UV protection
- Enabling efficient dyeing and finishing processes
- Minimizing waste generation

These technologies offer eco-friendly alternatives to traditional methods, decreasing environmental impact while improving textile performance.

40. Discuss the future of echo friendly fashion highlighting key trends impact s and challenges that shapes its role in the fashion industry.

A. The future of eco-friendly fashion is bright, with sustainability becoming a central theme in the fashion industry. Key trends shaping its role include:

- Circular Fashion: Designing clothes for reuse, recycling, and biodegradability, reducing waste and environmental footprint. Brands like Patagonia and Eileen Fisher are implementing take-back programs and repair services.
- Eco-Friendly Materials: Organic cotton, recycled polyester, hemp, and innovative materials like mycelium leather and Pinatex are gaining popularity.
- Slow Fashion: Emphasizing quality over quantity, encouraging consumers to invest in timeless pieces.
- Digital Fashion: Virtual try-ons, digital clothing, and AI-powered design tools reducing waste and environmental impact.

The impact of eco-friendly fashion is significant:
- Environmental Benefits: Reduced carbon emissions, water consumption, and waste generation.
- Social Benefits: Fair labor practices, safe working conditions, and transparent supply chains.
However, challenges remain:
- Scaling Sustainable Practices: Meeting mass production demands while maintaining eco-friendly standards.
- Consumer Awareness: Educating consumers about sustainable fashion and its benefits.
- Higher Costs: Sustainable materials and practices often increase prices, making eco-friendly fashion less accessible.
To overcome these challenges, collaboration between brands, governments, and consumers is crucial. Social media and influencers are playing a key role in promoting sustainable fashion awareness.

B. As consumers increasingly demand eco-friendly options, the fashion industry is expected to shift towards sustainability, making it the norm rather than the exception.

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